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|------|------------------------------------------------------------------------------------------------------|-----------|
|      |                                                                                                      |           |
| 5(a) | region where charge experiences an (electric) force                                                  | <b>B1</b> |
| 5(b) | graph: field strength zero from $x = 0$ to $x = R$                                                   | <b>B1</b> |
|      | curve with negative gradient, decreasing from $x = R$ to $x = 3R$                                    | <b>B1</b> |
|      | line passes through field strength $E$ at $x = R$ ,                                                  | <b>B1</b> |
|      | line passes through field strength $0.25E$ at $x = 2R$ <u>and</u> field strength $0.11E$ at $x = 3R$ | <b>B1</b> |

| Question | Answer                                                                           | Marks     |
|----------|----------------------------------------------------------------------------------|-----------|
| 5(c)     | field strength = $q / 4\pi\epsilon_0 x^2$                                        | <b>C1</b> |
|          | $2.0 \times 10^6 = q / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.26^2)$ | <b>C1</b> |
|          | $q = 1.5 \times 10^{-5} \text{ C}$                                               | <b>A1</b> |

| Question | Answer | Marks |
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